

100% Excellent!

PHY201 Exam on chapter(s): 7, 8

Name: _____

These questions assess your understanding of the material from Chapter(s) 7, 8 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Train cars are coupled together by being bumped into one another. Suppose two loaded train cars are moving toward one another, the first having a mass of 155000 kg and a velocity of 0.28 m/s, and the second having a mass of 152000 kg and a velocity of -0.26 m/s, where the minus sign indicates the direction of motion. What is the velocity of the two train cars after they couple together?

perfectly inelastic

$$m' = m_1 + m_2$$

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = \vec{v}' m'$$

$$\vec{v}' = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} = \frac{155000 \text{ kg} \cdot 0.28 \frac{\text{m}}{\text{s}} + 152000 \text{ kg} \cdot -0.26 \frac{\text{m}}{\text{s}}}{155000 \text{ kg} + 152000 \text{ kg}} = 1.26 \times 10^{-2} \frac{\text{m}}{\text{s}}$$

ANSWER: $1.26 \times 10^{-2} \frac{\text{m}}{\text{s}}$

2. Gabriel is playing baseball and is sliding into second base. He has a mass of 104.1 kg and begins sliding at a speed of 5.2 m/s. The average friction force acting against him during his motion is 610 N. What is the distance that he slides?

$$W_{\text{net}} = \Delta KE$$

$$F d \cos \theta = KE_f - KE_i$$

$$d = \frac{-\frac{1}{2} m v_i^2}{f \cos \theta} \quad \because KE_f = 0$$

$$d = \frac{-\frac{1}{2} \cdot 104.1 \text{ kg} \cdot 5.2^2 \frac{\text{m}^2}{\text{s}^2}}{610 \text{ N} \cos 180^\circ} = 2.307 \text{ m}$$

ANSWER: 2.307 m

3. Grace lifts a 97-kg barbell from the ground to a height of 2.2 m. How much work was done on the barbell?

$$W = F d \cos \theta$$

$$F = W = m g$$

$$W = m g d \cos \theta$$

$$W = 97 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 2.2 \text{ m} \cdot \cos 0^\circ = 2.09 \times 10^3 \text{ J}$$

ANSWER: $2.09 \times 10^3 \text{ J}$

+3

4. Sarah is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters a distance of 7.2 m to the top of a straight ramp. Sarah plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 1.1 m/s. If the height of the quarterpipe is 4.2 m, what must the height of the straight ramp be? Experimental studies in skate science show that the average friction force per mass on skaters as they skate across this particular surface is 1.06 N/kg.

$$W = Fd \cos \theta$$

$$\text{ANSWER: } 3.359 \text{ m}$$

$$W = m \frac{1.06 \text{ N}}{\text{kg}} \cdot 7.2 \text{ m} \cos 180^\circ$$

$$W = m \cdot \frac{-7.632 \text{ J}}{\text{kg}}$$

$$E_f = E_i + W_{nc}$$

$$KE_f + PE_f = PE_i + W_{nc}$$

$$\frac{1}{2} m v^2 + mgh_s = mgh_q + W_{nc}$$

$$\therefore KE_i = 0$$

$$h_s = \frac{gh_q + \frac{m \cdot \frac{-7.632 \text{ J}}{\text{kg}}}{m} - \frac{1}{2} v^2}{g}$$

$$h_s = \frac{9.8 \frac{\text{m}}{\text{s}^2} \cdot 4.2 \text{ m} - 7.632 \frac{\text{J}}{\text{kg}} - \frac{1}{2} 1.1^2 \frac{\text{m}^2}{\text{s}^2}}{9.8 \frac{\text{m}}{\text{s}^2}}$$

$$h_s = 3.359 \text{ m}$$

5. Ryan is a 62-kg athlete competing in a race. He begins from rest with an average acceleration of 3.0 m/s² for 2.6 s. What is his momentum at 2.6 s?

$$\vec{v} = \Delta \vec{v}$$

$$\vec{a} = \frac{\Delta \vec{v}}{t}$$

$$\vec{v} = \vec{a}t = 3.0 \frac{\text{m}}{\text{s}^2} \cdot 2.6 \text{ s} = 7.8 \frac{\text{m}}{\text{s}}$$

$$\vec{p} = m\vec{v} = 62 \text{ kg} \cdot 7.8 \frac{\text{m}}{\text{s}} = 483.6 \frac{\text{kg} \cdot \text{m}}{\text{s}}$$

$$\text{ANSWER: } 483.6 \frac{\text{kg} \cdot \text{m}}{\text{s}}$$

6. Brandon steps off a platform onto the floor from a height of 3.0 m. If Brandon has a mass of 71.6 kg and lands without bending his knees, his knee joints compress by 0.69 cm. Calculate the force acting on his knee joints. Note that since Brandon steps off the platform, his initial y-velocity is 0 m/s. Considering that 1000 N is roughly equal to 225 lbs, do you think this hurts Brandon's knees?

$$W = Fd \cos \theta$$

$$F = \frac{W}{d \cos \theta}$$

$$W = -\Delta PE$$

$$W = -PE_f + PE_i$$

$$W = mgh \quad \therefore PE_f = 0$$

$$\text{ANSWER: } 3.05 \times 10^5 \text{ N}$$

yes, it hurts

$$F = \frac{mgh}{d \cos \theta} = \frac{71.6 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 3.0 \text{ m}}{0.69 \times 10^{-2} \text{ m}} = 3.05 \times 10^5 \text{ N}$$

+3

7. Two identical billiard balls strike a rigid wall with the same speed and bounce without any change of speed. The first ball strikes perpendicular to the wall. The second ball strikes the wall at an angle of 5.5° from the perpendicular, and bounces off at an angle of 5.5° from perpendicular to the wall. Calculate the ratio of the magnitudes of impulses on the two balls from their collision with the wall.

$$\vec{J}_1 = \Delta \vec{p}_1 = \vec{p}_f - \vec{p}_i$$

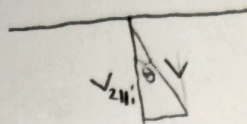
$$\vec{J}_1 = m(\vec{v}_f - \vec{v}_i)$$

$$\vec{J}_1 = m \cdot -2\vec{v}_i$$

$$\vec{J}_1 = -2m\vec{v}$$

$$\vec{v}_1 = -\vec{v}_i$$

$$\vec{v}_i = \vec{v}$$



$$v_{211i} = v \cos \theta$$

$$\vec{J}_2 = \Delta \vec{p}_2 = \vec{p}_{2f} - \vec{p}_{2i}$$

$$\vec{J}_2 = m(v_{211f} - v_{211i})$$

$$v_{211f} = -v_{211i}$$

$$\vec{J}_2 = m \cdot -2v_{211i}$$

$$\vec{J}_2 = -2m v \cos \theta$$

$$\text{ANSWER: } 1.0046$$

(ratio of 1 to 2)

8. Jessica places a 0.1-kg toy car next to a spring on a track. She moves the car backward to compress the spring 7.0 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Jessica greased the axles of the toy car, assume there is no friction. Jessica's previous experiments indicate that the spring has a force constant of 120 N/m.

$$E_1 = E_2$$

$$PE_{s1} = KE_2 \quad \because PE_{s2} = 0, KE_1 = 0$$

$$\frac{1}{2} k \Delta x^2 = \frac{1}{2} m v_2^2$$

$$v_2 = \Delta x \sqrt{\frac{k}{m}}$$

$$v_2 = 0.07 \text{ m} \sqrt{\frac{120 \text{ N/m}}{0.1 \text{ kg}}} = 2.425 \frac{\text{m}}{\text{s}}$$

$$\text{ANSWER: } 2.425 \frac{\text{m}}{\text{s}}$$

9. Scott is riding on a roller coaster, ascending a steep track as he approaches the top of a 89.7-m peak. Assuming Scott is moving at a speed of 3.6 m/s at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.

$$E_1 = E_2$$

$$PE_1 + KE_1 = PE_2 + KE_2$$

$$PE_1 + KE_1 = KE_2 \quad \because PE_2 = 0$$

$$mgh_1 + \frac{1}{2} m v_1^2 = \frac{1}{2} m v_2^2$$

$$v_2^2 = 2gh_1 + v_1^2$$

$$v_2 = \sqrt{2gh_1 + v_1^2}$$

$$v_2 = \sqrt{2 \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 89.7 \text{ m} + 3.6^2 \frac{\text{m}^2}{\text{s}^2}} = 42.08 \frac{\text{m}}{\text{s}}$$

$$\text{ANSWER: } 42.08 \frac{\text{m}}{\text{s}}$$

+3

10. A 52.2-kg block is resting on a so-called "heavy-duty" shelf at a height of 3.4 m above the ground. The shelf suddenly collapses and the block falls to the ground. What is the block's total mechanical energy at a height of 2.4 m, assuming negligible air resistance?

$$E_1 = E_2$$

ANSWER: $1.739 \times 10^3 \text{ J}$

$$E_1 = PE_1$$

$$E_2 = PE_1 = mgh_1 = 52.2 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 3.4 \text{ m} = 1.739 \times 10^3 \text{ J}$$

11. A 0.1-kg green box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary purple box, which has a mass of 0.1 kg. The green box emerges from the room at an angle of 31° north of east. The speed of the green box is originally 8.4 m/s and is 3.4 m/s after the collision. Calculate the speed of the purple box after the collision.

$$m_g = m_p$$

$$\Delta \vec{p}_x = 0$$

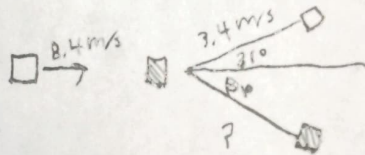
$$p_x = p'_x$$

$$m_g \vec{v}_{gx} + m_p \vec{v}_{px} = m_g \vec{v}'_{gx} + m_p \vec{v}'_{px}$$

$$m_g \vec{v}_{gx} = m_g \vec{v}'_{gx} + m_p \vec{v}'_{px} \quad \therefore \vec{v}_{px} = 0, m_p = m_g$$

$$\vec{v}_{gx} = \vec{v}'_{gx} + \vec{v}'_{px}$$

$$v_g = v'_g \cos \theta_g + v'_p \cos \theta_p$$



$$\vec{v}'_{gx} = v'_g \cos \theta_g$$

$$\vec{v}'_{gy} = v'_g \sin \theta_g$$

$$\vec{v}'_{px} = v'_p \cos \theta_p$$

$$\vec{v}'_{py} = -v'_p \sin \theta_p$$

$$\vec{v}_{gx} = v_g$$

$$\vec{v}_{gy} = 0$$

$$\vec{v}_p = 0$$

ANSWER: $5.76 \frac{\text{m}}{\text{s}}$

$$\Delta \vec{p}_y = 0$$

$$p_y = p'_y$$

$$m_g \vec{v}_{gy} + m_p \vec{v}_{py} = m_g \vec{v}'_{gy} + m_p \vec{v}'_{py}$$

$$0 = m_g \vec{v}'_{gy} + m_p \vec{v}'_{py} \quad \begin{cases} \vec{v}'_{gy} = 0 \\ \vec{v}'_{py} = 0 \\ m_p = m_g \end{cases}$$

$$\vec{v}'_{py} = -\vec{v}'_{gy}$$

12. A rocket with a mass at liftoff of 2.7 million kg burns fuel at a rate of 15800 kg/s. Its exhaust velocity was measured to be 3100 m/s. Calculate its acceleration at liftoff.

$$v'_p \cos \theta_p = v_g - v'_g \cos \theta_g$$

$$\vec{v}'_{py} = -v'_p \sin \theta_p$$

$$\frac{-\vec{v}'_{py} \cos \theta_p}{\sin \theta_p} = v_g - v'_g \cos \theta_g$$

$$v'_p = \frac{-\vec{v}'_{py}}{\sin \theta_p}$$

$$\vec{v}'_{gy} \cot \theta_p = v_g - v'_g \cos \theta_g \quad \left\{ \begin{array}{l} v'_p = -v'_g \\ v'_g = -v'_g \end{array} \right.$$

$$\frac{v'_g \sin \theta_g}{v_g - v'_g \cos \theta_g} = \tan \theta_p$$

$$\theta_p = \tan^{-1} \left(\frac{3.4 \sin 31^\circ}{8.4 - 3.4 \cos 31^\circ} \right) = 17.7^\circ$$

$$v'_p = \frac{-\vec{v}'_{py}}{\sin \theta_p}$$

$$v'_p = \frac{\vec{v}'_{gy}}{\sin \theta_p}, \quad v'_p = \frac{v'_g \sin \theta_g}{\sin \theta_p} = \frac{3.4 \sin 31^\circ}{\sin 17.7^\circ} = 5.76 \frac{\text{m}}{\text{s}}$$

ANSWER: $8.34 \frac{\text{m}}{\text{s}^2}$

$$a_r = \frac{v_b}{m_i} \frac{\Delta m}{\Delta t} - g$$

$$a_r = \frac{3100 \frac{\text{m}}{\text{s}}}{2.7 \times 10^6 \text{ kg}} \cdot \frac{15800 \text{ kg}}{\text{s}} - 9.8 \frac{\text{m}}{\text{s}^2} = 8.34 \frac{\text{m}}{\text{s}^2}$$

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1. 1.3 cm/s
2. 2.307 m
3. 2091 J
4. 3.36 m
5. 484 kg·m/s
6. 305 kN
7. 1.0
8. 242 cm/s
9. 42.1 m/s
10. 1739 J
11. 5.76 m/s
12. 8.3 m/s²